OS MINIPROJECT

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#1 Multi-container supervision

```
sparsha@sparsha-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine start alpha
 ../rootfs-alpha /bin/sh
sudo ./engine start beta ../rootfs-beta /bin/sh
[sudo] password for sparsha:
Request sent to supervisor
Request sent to supervisor
sparsha@sparsha-VirtualBox:~/OS-Jackfruit/boilerplate$

sparsha@sparsha-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine supervisor .
[sudo] password for sparsha:
Supervisor running...
Received request: alpha
Started container alpha with PID 8172
/ # Received request: beta
Started container beta with PID 8179
/ #
```

Figure 1: Single supervisor process managing multiple isolated containers concurrently using namespace-based isolation.

#2 Metadata tracking

```
sparsha@sparsha-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine supervisor ../rootfs-base
[sudo] password for sparsha:
✅ Supervisor listening on /tmp/mini_runtime.sock
DEBUG → kind=1 id=alpha
🚀 Started container alpha (PID=3060)
DEBUG → kind=3 id=

ID      PID     STATE
alpha   3060    running

DEBUG → kind=5 id=alpha
🛑 Stopped container alpha
cpu_hog alive elapsed=1 accumulator=9916004950501539656
cpu_hog alive elapsed=2 accumulator=5411280940047574187
cpu_hog alive elapsed=3 accumulator=13309162321682427752
cpu_hog alive elapsed=4 accumulator=3400232458366859699
cpu_hog alive elapsed=5 accumulator=10636221923858737800
cpu_hog alive elapsed=6 accumulator=6949234214139522945
cpu_hog alive elapsed=7 accumulator=8383196047603593706
cpu_hog alive elapsed=8 accumulator=3710644847272149191
cpu_hog alive elapsed=9 accumulator=16254265531719119043
cpu_hog alive elapsed=10 accumulator=3324318866112591382
cpu_hog done duration=10 accumulator=3324318866112591382
```

Figure 2: Supervisor metadata table displaying tracked container IDs, PIDs, and runtime states.

#3 Bounded-buffer logging

```
sparsha@sparsha-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine start alpha ../rootfs-alpha "/cpu_hog
"
[sudo] password for sparsha:
sparsha@sparsha-VirtualBox:~/OS-Jackfruit/boilerplate$ cat alpha.log
cpu_hog alive elapsed=1 accumulator=8209518881885477365
cpu_hog alive elapsed=2 accumulator=15874328205650978642
cpu_hog alive elapsed=3 accumulator=5016846799768498456
cpu_hog alive elapsed=4 accumulator=17774332290597364650
cpu_hog alive elapsed=5 accumulator=17534359882983321838
cpu_hog alive elapsed=6 accumulator=17150532441505844002
cpu_hog alive elapsed=7 accumulator=7969152635124534770
cpu_hog alive elapsed=8 accumulator=1885582551159771952
cpu_hog alive elapsed=9 accumulator=5171162874694040453
cpu_hog alive elapsed=10 accumulator=8647258464233328018
cpu_hog done duration=10 accumulator=8647258464233328018
sparsha@sparsha-VirtualBox:~/OS-Jackfruit/boilerplate$
```

```
sparsha@sparsha-VirtualBox:~/OS-Jackfruit/boilerplate$ cat alpha.log
allocation=1 chunk=8MB total=8MB
allocation=49 chunk=8MB total=392MB
allocation=2 chunk=8MB total=16MB
allocation=50 chunk=8MB total=400MB
allocation=3 chunk=8MB total=24MB
allocation=51 chunk=8MB total=408MB
allocation=4 chunk=8MB total=32MB
allocation=52 chunk=8MB total=416MB
allocation=5 chunk=8MB total=40MB
allocation=53 chunk=8MB total=424MB
allocation=6 chunk=8MB total=48MB
allocation=54 chunk=8MB total=432MB
```

Figure 3: Container stdout captured through the producer-consumer logging pipeline into persistent per-container log files.

#4 CLI and IPC

```
sparsha@sparsha-VirtualBox:~$ cd ~/OS-Jackfruit/boilerplate
sparsha@sparsha-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine start alpha
 ../rootfs-alpha /bin/sh
Request sent to supervisor
sparsha@sparsha-VirtualBox:~/OS-Jackfruit/boilerplate$
```

```
sparsha@sparsha-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine supervisor ../rootfs-base
[sudo] password for sparsha:
Supervisor running...
Received request: alpha
```

Figure 4: CLI command transmitted to the supervisor through UNIX domain socket IPC.

#5 Soft-limit warning & #6 Hard-limit enforcement

```
sparsha@sparsha-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo dmesg | tail -20
[  528.689699] audit: type=1400 audit(1776881220.456:76): apparmor="STATUS" operation="profile_replace" profile="
unconfined" name="snap.firefox.hook.install" pid=2794 comm="apparmor_parser"
[  529.752229] audit: type=1400 audit(1776881221.707:77): apparmor="STATUS" operation="profile_replace" profile="
unconfined" name="snap.firefox.hook.post-refresh" pid=2795 comm="apparmor_parser"
[  533.482501] audit: type=1400 audit(1776881225.425:78): apparmor="STATUS" operation="profile_replace" profile="
unconfined" name="snap.firefox.geckodriver" pid=2791 comm="apparmor_parser"
[  533.611640] audit: type=1400 audit(1776881225.559:79): apparmor="STATUS" operation="profile_replace" profile="
unconfined" name="snap.firefox.firefox" pid=2790 comm="apparmor_parser"
[  534.281335] audit: type=1400 audit(1776881226.235:80): apparmor="STATUS" operation="profile_replace" profile="
unconfined" name="snap-update-ns.firefox" pid=2789 comm="apparmor_parser"
[  534.310104] audit: type=1400 audit(1776881226.264:81): apparmor="STATUS" operation="profile_replace" info="sam
e as current profile, skipping" profile="unconfined" name="snap-update-ns.mesa-2404" pid=2800 comm="apparmor_pars
er"
[  534.319462] audit: type=1400 audit(1776881226.272:82): apparmor="STATUS" operation="profile_replace" info="sam
e as current profile, skipping" profile="unconfined" name="snap.mesa-2404.hook.connect-plug-kernel-gpu-2404" pid=
2802 comm="apparmor_parser"
[  534.319481] audit: type=1400 audit(1776881226.275:83): apparmor="STATUS" operation="profile_replace" info="sam
e as current profile, skipping" profile="unconfined" name="snap.mesa-2404.hook.disconnect-plug-kernel-gpu-2404" p
id=2803 comm="apparmor_parser"
[  534.319487] audit: type=1400 audit(1776881226.275:84): apparmor="STATUS" operation="profile_replace" info="sam
e as current profile, skipping" profile="unconfined" name="snap.mesa-2404.hook.install" pid=2804 comm="apparmor_p
arser"
[  534.324304] audit: type=1400 audit(1776881226.277:85): apparmor="STATUS" operation="profile_replace" info="sam
e as current profile, skipping" profile="unconfined" name="snap.mesa-2404.component-monitor" pid=2801 comm="appar
mor_parser"
[  534.324319] audit: type=1400 audit(1776881226.278:86): apparmor="STATUS" operation="profile_replace" info="sam
e as current profile, skipping" profile="unconfined" name="snap.mesa-2404.hook.post-refresh" pid=2805 comm="appar
mor_parser"
[  535.243499] audit: type=1400 audit(1776881227.195:87): apparmor="DENIED" operation="capable" class="cap" profi
le="/usr/lib/snapd/snap-confine" pid=2808 comm="snap-confine" capability=12  capname="net_admin"
[  539.906673] kauditd_printk_skb: 1 callbacks suppressed
[  539.906681] audit: type=1400 audit(1776881231.852:89): apparmor="DENIED" operation="open" class="file" profile
="snap-update-ns.firefox" name="/usr/local/share/" pid=2830 comm="snap-update-ns" requested_mask="r" denied_mask=
"r" fsuid=0 ouid=0
[  794.684428] [container_monitor] Module loaded. Device: /dev/container_monitor
[  805.586067] [container_monitor] Module unloaded.
[  805.710926] [container_monitor] Module loaded. Device: /dev/container_monitor
[  838.999856] [container_monitor] Registering container=alpha pid=3295 soft=52428800 hard=83886080
[  846.303631] [container_monitor] SOFT LIMIT container=alpha pid=3295 rss=59244544 limit=52428800
[  849.523144] [container_monitor] HARD LIMIT container=alpha pid=3295 rss=84410368 limit=83886080
sparsha@sparsha-VirtualBox:~/OS-Jackfruit/boilerplate$
```

Figure 5: Kernel memory monitor detecting soft-limit and hard-limit memory violations for a running container.

#7 Scheduling experiment

```
sparsha@sparsha-VirtualBox:~/OS-Jackfruit/boilerplate$ cat alpha.log
allocation=1 chunk=8MB total=8MB
allocation=2 chunk=8MB total=16MB
allocation=3 chunk=8MB total=24MB
allocation=4 chunk=8MB total=32MB
allocation=5 chunk=8MB total=40MB
allocation=6 chunk=8MB total=48MB
allocation=7 chunk=8MB total=56MB
allocation=8 chunk=8MB total=64MB
allocation=9 chunk=8MB total=72MB
cpu_hog alive elapsed=1 accumulator=15185874102498456079
cpu_hog alive elapsed=2 accumulator=15134780582755709992
cpu_hog alive elapsed=3 accumulator=12816727868886622546
cpu_hog alive elapsed=4 accumulator=6763765253710130082
cpu_hog alive elapsed=5 accumulator=8903470123451299616
cpu_hog alive elapsed=6 accumulator=5375079760232783586
cpu_hog alive elapsed=7 accumulator=16812568854142740162
cpu_hog alive elapsed=8 accumulator=17449204671921125605
cpu_hog alive elapsed=9 accumulator=18185809936328222330
cpu_hog done duration=10 accumulator=8958993646004429486
cpu_hog alive elapsed=1 accumulator=16502406212220047608
cpu_hog alive elapsed=2 accumulator=17282675927911010315
cpu_hog alive elapsed=3 accumulator=2407176993267970583
cpu_hog alive elapsed=4 accumulator=14470743870057291994
cpu_hog alive elapsed=5 accumulator=2407051457689888279
cpu_hog alive elapsed=6 accumulator=15840754873865110003
cpu_hog alive elapsed=7 accumulator=5949572446503855355
cpu_hog alive elapsed=8 accumulator=11507296620158810215
cpu_hog alive elapsed=9 accumulator=16160146074774733078
cpu_hog done duration=10 accumulator=1026734693778742279
```

```
sparsha@sparsha-VirtualBox:~/OS-Jackfruit/boilerplate$ cat beta.log
cpu_hog alive elapsed=1 accumulator=10081497582719097488
cpu_hog alive elapsed=2 accumulator=10651648264269416423
cpu_hog alive elapsed=3 accumulator=10878699806425003992
cpu_hog alive elapsed=4 accumulator=615426360032193715
cpu_hog alive elapsed=5 accumulator=10073694147306290513
cpu_hog alive elapsed=6 accumulator=742140081150679467
cpu_hog alive elapsed=7 accumulator=9106433764019401660
cpu_hog alive elapsed=8 accumulator=1456591223275704500
cpu_hog alive elapsed=9 accumulator=16121723613267152258
cpu_hog alive elapsed=10 accumulator=9529846226719356113
cpu_hog done duration=10 accumulator=9529846226719356113
cpu_hog alive elapsed=1 accumulator=13744405765080408863
cpu_hog alive elapsed=2 accumulator=9969967254363722279
cpu_hog alive elapsed=3 accumulator=11551110210843972222
cpu_hog alive elapsed=4 accumulator=2713845903893197045
cpu_hog alive elapsed=5 accumulator=13725691898947803627
cpu_hog alive elapsed=6 accumulator=10470275290410112395
cpu_hog alive elapsed=7 accumulator=16586448037993072381
cpu_hog alive elapsed=8 accumulator=12792637604154532570
cpu_hog alive elapsed=9 accumulator=10857844652108803461
cpu_hog done duration=10 accumulator=7236156417249500509
sparsha@sparsha-VirtualBox:~/OS-Jackfruit/boilerplate$
```

Figure 6: Scheduling experiment comparing concurrent CPU-bound workloads running with different scheduling priorities. The experiment used two CPU-bound workloads with different nice values. The higher-priority workload received more favorable CPU scheduling from the Linux Completely Fair Scheduler (CFS), demonstrating how Linux dynamically distributes CPU time based on process priority.

#8 Clean teardown

```
sparsha@sparsha-VirtualBox:~/OS-Jackfruit/boilerplate$ ps aux | grep defunct
sparsha     3112  0.0  0.0   9080  2572 pts/2    R+   01:46   0:00 grep --color=auto defunct
sparsha@sparsha-VirtualBox:~/OS-Jackfruit/boilerplate$
```

Figure 7: Clean teardown verification showing that no zombie or defunct container processes remain after shutdown.